# Analysis report

Holzinger–Swineford · MLR · listwise · N = 156 + 145 · lavaan 0.7.3

Table 1: Parameter estimates

| Group | Left | Relation | Right | Label / value | Estimate | SE | z | p | 95% CI lower | 95% CI upper | std.all |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Pasteur | visual | =~ | x1 | | 1.000 | 0.000 | — | — | 1.000 | 1.000 | 0.887 |
| Pasteur | visual | =~ | x2 | | 0.394 | 0.197 | 1.996 | .046 | 0.007 | 0.780 | 0.336 |
| Pasteur | visual | =~ | x3 | | 0.570 | 0.201 | 2.838 | .005 | 0.176 | 0.964 | 0.515 |
| Pasteur | textual | =~ | x4 | | 1.000 | 0.000 | — | — | 1.000 | 1.000 | 0.823 |
| Pasteur | textual | =~ | x5 | | 1.183 | 0.106 | 11.213 | < .001 | 0.977 | 1.390 | 0.856 |
| Pasteur | textual | =~ | x6 | | 0.875 | 0.093 | 9.388 | < .001 | 0.692 | 1.058 | 0.838 |
| Pasteur | speed | =~ | x7 | | 1.000 | 0.000 | — | — | 1.000 | 1.000 | 0.547 |
| Pasteur | speed | =~ | x8 | | 1.125 | 0.262 | 4.298 | < .001 | 0.612 | 1.638 | 0.682 |
| Pasteur | speed | =~ | x9 | | 0.922 | 0.297 | 3.099 | .002 | 0.339 | 1.505 | 0.551 |
| Pasteur | x1 | ~~ | x1 | | 0.298 | 0.338 | 0.882 | .378 | -0.365 | 0.961 | 0.214 |
| Pasteur | x2 | ~~ | x2 | | 1.334 | 0.179 | 7.441 | < .001 | 0.982 | 1.685 | 0.887 |
| Pasteur | x3 | ~~ | x3 | | 0.989 | 0.154 | 6.444 | < .001 | 0.689 | 1.290 | 0.735 |
| Pasteur | x4 | ~~ | x4 | | 0.425 | 0.073 | 5.867 | < .001 | 0.283 | 0.567 | 0.322 |
| Pasteur | x5 | ~~ | x5 | | 0.456 | 0.086 | 5.304 | < .001 | 0.287 | 0.624 | 0.267 |
| Pasteur | x6 | ~~ | x6 | | 0.290 | 0.055 | 5.267 | < .001 | 0.182 | 0.398 | 0.297 |
| Pasteur | x7 | ~~ | x7 | | 0.820 | 0.132 | 6.211 | < .001 | 0.561 | 1.079 | 0.701 |
| Pasteur | x8 | ~~ | x8 | | 0.510 | 0.101 | 5.048 | < .001 | 0.312 | 0.708 | 0.535 |
| Pasteur | x9 | ~~ | x9 | | 0.680 | 0.126 | 5.406 | < .001 | 0.434 | 0.927 | 0.696 |
| Pasteur | visual | ~~ | visual | | 1.097 | 0.378 | 2.899 | .004 | 0.355 | 1.839 | 1.000 |
| Pasteur | textual | ~~ | textual | | 0.894 | 0.160 | 5.577 | < .001 | 0.580 | 1.209 | 1.000 |
| Pasteur | speed | ~~ | speed | | 0.350 | 0.140 | 2.502 | .012 | 0.076 | 0.623 | 1.000 |
| Pasteur | visual | ~~ | textual | | 0.479 | 0.144 | 3.334 | < .001 | 0.197 | 0.761 | 0.484 |
| Pasteur | visual | ~~ | speed | | 0.185 | 0.076 | 2.449 | .014 | 0.037 | 0.333 | 0.299 |
| Pasteur | textual | ~~ | speed | | 0.182 | 0.076 | 2.402 | .016 | 0.033 | 0.330 | 0.325 |
| Pasteur | x1 | ~1 | | | 4.941 | 0.095 | 52.249 | < .001 | 4.756 | 5.127 | 4.183 |
| Pasteur | x2 | ~1 | | | 5.984 | 0.098 | 60.949 | < .001 | 5.792 | 6.176 | 4.880 |
| Pasteur | x3 | ~1 | | | 2.487 | 0.093 | 26.778 | < .001 | 2.305 | 2.669 | 2.144 |
| Pasteur | x4 | ~1 | | | 2.823 | 0.092 | 30.689 | < .001 | 2.642 | 3.003 | 2.457 |
| Pasteur | x5 | ~1 | | | 3.995 | 0.105 | 38.183 | < .001 | 3.790 | 4.200 | 3.057 |
| Pasteur | x6 | ~1 | | | 1.922 | 0.079 | 24.321 | < .001 | 1.767 | 2.077 | 1.947 |
| Pasteur | x7 | ~1 | | | 4.432 | 0.087 | 51.181 | < .001 | 4.263 | 4.602 | 4.098 |
| Pasteur | x8 | ~1 | | | 5.563 | 0.078 | 71.214 | < .001 | 5.410 | 5.716 | 5.702 |
| Pasteur | x9 | ~1 | | | 5.418 | 0.079 | 68.440 | < .001 | 5.263 | 5.573 | 5.480 |
| Pasteur | visual | ~1 | | | 0.000 | 0.000 | — | — | 0.000 | 0.000 | 0.000 |
| Pasteur | textual | ~1 | | | 0.000 | 0.000 | — | — | 0.000 | 0.000 | 0.000 |

Table 1 (Parameter estimates)

| Group | Left | Relation | Right | Label / value | Estimate | SE | z | p | 95% CI lower | 95% CI upper | std.all |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Pasteur | speed | ~1 | | | 0.000 | 0.000 | — | — | 0.000 | 0.000 | 0.000 |
| Grant-White | visual | =~ | x1 | | 1.000 | 0.000 | — | — | 1.000 | 1.000 | 0.677 |
| Grant-White | visual | =~ | x2 | | 0.736 | 0.190 | 3.870 | < .001 | 0.363 | 1.109 | 0.517 |
| Grant-White | visual | =~ | x3 | | 0.925 | 0.216 | 4.281 | < .001 | 0.501 | 1.348 | 0.694 |
| Grant-White | textual | =~ | x4 | | 1.000 | 0.000 | — | — | 1.000 | 1.000 | 0.866 |
| Grant-White | textual | =~ | x5 | | 0.990 | 0.089 | 11.179 | < .001 | 0.816 | 1.163 | 0.829 |
| Grant-White | textual | =~ | x6 | | 0.963 | 0.091 | 10.588 | < .001 | 0.785 | 1.142 | 0.826 |
| Grant-White | speed | =~ | x7 | | 1.000 | 0.000 | — | — | 1.000 | 1.000 | 0.659 |
| Grant-White | speed | =~ | x8 | | 1.226 | 0.162 | 7.560 | < .001 | 0.908 | 1.544 | 0.796 |
| Grant-White | speed | =~ | x9 | | 1.058 | 0.250 | 4.227 | < .001 | 0.567 | 1.548 | 0.701 |
| Grant-White | x1 | ~~ | x1 | | 0.715 | 0.182 | 3.938 | < .001 | 0.359 | 1.071 | 0.542 |
| Grant-White | x2 | ~~ | x2 | | 0.899 | 0.142 | 6.338 | < .001 | 0.621 | 1.177 | 0.733 |
| Grant-White | x3 | ~~ | x3 | | 0.557 | 0.122 | 4.568 | < .001 | 0.318 | 0.796 | 0.519 |
| Grant-White | x4 | ~~ | x4 | | 0.315 | 0.066 | 4.764 | < .001 | 0.186 | 0.445 | 0.251 |
| Grant-White | x5 | ~~ | x5 | | 0.419 | 0.071 | 5.876 | < .001 | 0.279 | 0.559 | 0.312 |
| Grant-White | x6 | ~~ | x6 | | 0.406 | 0.076 | 5.363 | < .001 | 0.258 | 0.554 | 0.317 |
| Grant-White | x7 | ~~ | x7 | | 0.600 | 0.100 | 5.995 | < .001 | 0.404 | 0.797 | 0.566 |
| Grant-White | x8 | ~~ | x8 | | 0.401 | 0.159 | 2.516 | .012 | 0.089 | 0.714 | 0.367 |
| Grant-White | x9 | ~~ | x9 | | 0.535 | 0.140 | 3.814 | < .001 | 0.260 | 0.810 | 0.509 |
| Grant-White | visual | ~~ | visual | | 0.604 | 0.194 | 3.105 | .002 | 0.223 | 0.985 | 1.000 |
| Grant-White | textual | ~~ | textual | | 0.942 | 0.162 | 5.807 | < .001 | 0.624 | 1.260 | 1.000 |
| Grant-White | speed | ~~ | speed | | 0.461 | 0.120 | 3.839 | < .001 | 0.226 | 0.697 | 1.000 |
| Grant-White | visual | ~~ | textual | | 0.408 | 0.115 | 3.535 | < .001 | 0.182 | 0.634 | 0.541 |
| Grant-White | visual | ~~ | speed | | 0.276 | 0.101 | 2.731 | .006 | 0.078 | 0.474 | 0.523 |
| Grant-White | textual | ~~ | speed | | 0.222 | 0.105 | 2.115 | .034 | 0.016 | 0.427 | 0.336 |
| Grant-White | x1 | ~1 | | | 4.930 | 0.095 | 51.696 | < .001 | 4.743 | 5.117 | 4.293 |
| Grant-White | x2 | ~1 | | | 6.200 | 0.092 | 67.416 | < .001 | 6.020 | 6.380 | 5.599 |
| Grant-White | x3 | ~1 | | | 1.996 | 0.086 | 23.195 | < .001 | 1.827 | 2.164 | 1.926 |
| Grant-White | x4 | ~1 | | | 3.317 | 0.093 | 35.625 | < .001 | 3.135 | 3.500 | 2.959 |
| Grant-White | x5 | ~1 | | | 4.712 | 0.096 | 48.986 | < .001 | 4.524 | 4.901 | 4.068 |
| Grant-White | x6 | ~1 | | | 2.469 | 0.094 | 26.277 | < .001 | 2.285 | 2.653 | 2.182 |
| Grant-White | x7 | ~1 | | | 3.921 | 0.086 | 45.819 | < .001 | 3.753 | 4.089 | 3.805 |
| Grant-White | x8 | ~1 | | | 5.488 | 0.087 | 63.174 | < .001 | 5.318 | 5.659 | 5.246 |
| Grant-White | x9 | ~1 | | | 5.327 | 0.085 | 62.571 | < .001 | 5.160 | 5.494 | 5.196 |
| Grant-White | visual | ~1 | | | 0.000 | 0.000 | — | — | 0.000 | 0.000 | 0.000 |
| Grant-White | textual | ~1 | | | 0.000 | 0.000 | — | — | 0.000 | 0.000 | 0.000 |

Table 1 (Parameter estimates)

| Group | Left | Relation | Right | Label / value | Estimate | SE | z | p | 95% CI lower | 95% CI upper | std.all |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Grant-White | speed | ~1 | | | 0.000 | 0.000 | — | — | 0.000 | 0.000 | 0.000 |

Note. Estimates and intervals are unstandardised; Std. all is fully standardised. Values are rounded only for display. A dash indicates unavailable output. 95% CI.

Table 2: Fit measures

| Measure | Value |
|---|---|
| $\chi^2$ | 115.851 |
| df | 48.000 |
| p ($\chi^2$) | < .001 |
| CFI | 0.923 |
| TLI | 0.885 |
| RMSEA | 0.097 |
| RMSEA 90% CI lower | 0.075 |
| RMSEA 90% CI upper | 0.120 |
| SRMR | 0.068 |
| Scaled $\chi^2$ | 121.741 |
| Scaled df | 48.000 |
| p (scaled $\chi^2$) | < .001 |
| Robust CFI | 0.921 |
| Robust TLI | 0.881 |
| Robust RMSEA | 0.099 |
| Robust RMSEA 90% CI lower | 0.077 |
| Robust RMSEA 90% CI upper | 0.120 |
| Scaled CFI | 0.914 |
| Scaled TLI | 0.872 |
| Scaled RMSEA | 0.101 |
| Scaled RMSEA 90% CI lower | 0.078 |
| Scaled RMSEA 90% CI upper | 0.124 |

Note. Scaled and robust statistics are labelled explicitly. Compare the appropriate version for your estimator.

Table 3: Explained variance ($R^2$)

| Group | Left | $R^2$ |
|---|---|---|
| Pasteur | x1 | 0.786 |
| Pasteur | x2 | 0.113 |
| Pasteur | x3 | 0.265 |
| Pasteur | x4 | 0.678 |
| Pasteur | x5 | 0.733 |
| Pasteur | x6 | 0.703 |
| Pasteur | x7 | 0.299 |
| Pasteur | x8 | 0.465 |
| Pasteur | x9 | 0.304 |
| Grant-White | x1 | 0.458 |
| Grant-White | x2 | 0.267 |
| Grant-White | x3 | 0.481 |
| Grant-White | x4 | 0.749 |
| Grant-White | x5 | 0.688 |
| Grant-White | x6 | 0.683 |
| Grant-White | x7 | 0.434 |
| Grant-White | x8 | 0.633 |
| Grant-White | x9 | 0.491 |

Table 4: Measurement invariance

| Stage | $\chi^2$ | df | p | CFI | RMSEA | SRMR | ΔCFI | ΔRMSEA | $\Delta\chi^2$ | Δdf | Difference p | CFI variant | RMSEA variant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Configural | 121.741 | 48 | < .001 | 0.921 | 0.099 | 0.068 | — | — | — | — | — | cfi.robust | rmsea.robust |
| Metric | 125.997 | 54 | < .001 | 0.920 | 0.093 | 0.072 | -0.001 | -0.005 | 6.567 | 6 | .363 | cfi.robust | rmsea.robust |
| Scalar | 166.748 | 60 | < .001 | 0.881 | 0.108 | 0.082 | -0.039 | 0.014 | 40.838 | 6 | < .001 | cfi.robust | rmsea.robust |
| Strict | 181.429 | 69 | < .001 | 0.873 | 0.104 | 0.088 | -0.008 | -0.004 | 15.694 | 9 | .074 | cfi.robust | rmsea.robust |

Note. Each stage is compared with the previous stage using lavaan's difference test. ΔCFI and ΔRMSEA are signed changes; the exact fit-index variants are reported. These statistics do not automatically establish invariance. Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001") Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001") Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

Table 5: Modification indices

| Group | Left | Relation | Right | MI | EPC | Standardised EPC |
|---|---|---|---|---|---|---|
| Grant-White | x7 | ~~ | x8 | 24.819 | 0.612 | 1.247 |
| Grant-White | visual | =~ | x9 | 24.539 | 0.748 | 0.567 |
| Pasteur | x4 | ~~ | x6 | 11.280 | -0.326 | -0.928 |
| Grant-White | visual | =~ | x7 | 11.267 | -0.504 | -0.380 |
| Pasteur | visual | =~ | x9 | 11.073 | 0.304 | 0.322 |
| Pasteur | textual | =~ | x1 | 10.185 | 0.944 | 0.756 |
| Pasteur | visual | =~ | x5 | 9.593 | -0.281 | -0.225 |
| Grant-White | textual | =~ | x8 | 9.572 | -0.290 | -0.269 |
| Grant-White | x1 | ~~ | x9 | 9.332 | 0.205 | 0.332 |
| Grant-White | textual | =~ | x9 | 9.201 | 0.268 | 0.253 |
| Pasteur | x2 | ~~ | x3 | 8.961 | 0.329 | 0.286 |
| Grant-White | x8 | ~~ | x9 | 8.494 | -0.398 | -0.859 |
| Pasteur | x4 | ~~ | x7 | 7.197 | 0.159 | 0.270 |
| Pasteur | x2 | ~~ | x7 | 6.851 | -0.241 | -0.230 |
| Pasteur | textual | =~ | x3 | 6.203 | -0.393 | -0.320 |
| Pasteur | x4 | ~~ | x5 | 5.463 | 0.315 | 0.715 |
| Grant-White | x1 | ~~ | x7 | 5.236 | -0.158 | -0.241 |
| Pasteur | visual | =~ | x7 | 5.182 | -0.227 | -0.220 |
| Pasteur | x3 | ~~ | x9 | 4.526 | 0.158 | 0.192 |
| Grant-White | x7 | ~~ | x9 | 4.332 | -0.208 | -0.367 |
| Grant-White | x4 | ~~ | x8 | 4.201 | -0.095 | -0.268 |
| Pasteur | x7 | ~~ | x8 | 4.187 | 0.342 | 0.528 |
| Pasteur | visual | =~ | x6 | 3.972 | 0.137 | 0.146 |
| Pasteur | x1 | ~~ | x2 | 3.971 | -0.340 | -0.539 |
| Pasteur | x1 | ~~ | x4 | 3.932 | 0.119 | 0.334 |
| Pasteur | x3 | ~~ | x5 | 3.757 | -0.134 | -0.199 |
| Grant-White | x3 | ~~ | x7 | 3.534 | -0.116 | -0.201 |
| Grant-White | x2 | ~~ | x3 | 3.441 | 0.176 | 0.249 |
| Pasteur | speed | =~ | x5 | 3.333 | -0.293 | -0.132 |
| Pasteur | x2 | ~~ | x9 | 2.830 | 0.141 | 0.148 |
| Grant-White | speed | =~ | x1 | 2.780 | 0.361 | 0.214 |
| Grant-White | visual | =~ | x8 | 2.749 | -0.273 | -0.203 |
| Pasteur | speed | =~ | x6 | 2.661 | 0.200 | 0.120 |
| Grant-White | speed | =~ | x5 | 2.579 | 0.189 | 0.111 |
| Pasteur | x2 | ~~ | x4 | 2.275 | -0.107 | -0.141 |
| Pasteur | x1 | ~~ | x5 | 2.240 | -0.100 | -0.271 |
| Pasteur | x5 | ~~ | x7 | 1.914 | -0.090 | -0.147 |
| Pasteur | x3 | ~~ | x7 | 1.724 | -0.107 | -0.118 |
| Pasteur | x1 | ~~ | x9 | 1.595 | 0.091 | 0.203 |
| Pasteur | x6 | ~~ | x8 | 1.566 | 0.055 | 0.143 |
| Pasteur | x7 | ~~ | x9 | 1.543 | -0.157 | -0.210 |
| Pasteur | visual | =~ | x4 | 1.498 | 0.099 | 0.090 |
| Grant-White | x1 | ~~ | x3 | 1.468 | -0.164 | -0.260 |
| Pasteur | x4 | ~~ | x8 | 1.349 | -0.061 | -0.130 |
| Pasteur | x3 | ~~ | x6 | 1.345 | 0.062 | 0.115 |
| Grant-White | x3 | ~~ | x6 | 1.314 | 0.061 | 0.129 |
| Pasteur | x5 | ~~ | x6 | 1.303 | 0.138 | 0.380 |
| Pasteur | x4 | ~~ | x9 | 1.290 | -0.062 | -0.114 |
| Pasteur | textual | =~ | x8 | 1.176 | -0.120 | -0.116 |
| Grant-White | x2 | ~~ | x5 | 1.146 | -0.067 | -0.109 |
| Grant-White | speed | =~ | x3 | 1.071 | -0.206 | -0.135 |

Table 5 (Modification indices)

| Group | Left | Relation | Right | MI | EPC | Standardised EPC |
|---|---|---|---|---|---|---|
| Grant-White | x3 | ~~ | x9 | 1.047 | 0.062 | 0.113 |
| Pasteur | x2 | ~~ | x6 | 1.007 | 0.060 | 0.096 |
| Pasteur | visual | =~ | x8 | 0.925 | -0.097 | -0.104 |
| Pasteur | x1 | ~~ | x7 | 0.804 | -0.071 | -0.143 |
| Grant-White | textual | =~ | x2 | 0.741 | -0.111 | -0.098 |
| Grant-White | x3 | ~~ | x5 | 0.699 | -0.046 | -0.095 |
| Grant-White | speed | =~ | x4 | 0.691 | -0.093 | -0.056 |
| Grant-White | x5 | ~~ | x7 | 0.687 | 0.044 | 0.087 |
| Pasteur | x1 | ~~ | x3 | 0.641 | -0.240 | -0.442 |
| Pasteur | x1 | ~~ | x8 | 0.628 | -0.059 | -0.150 |
| Grant-White | x5 | ~~ | x9 | 0.587 | 0.039 | 0.083 |
| Pasteur | speed | =~ | x2 | 0.587 | -0.168 | -0.081 |
| Grant-White | speed | =~ | x2 | 0.569 | -0.145 | -0.089 |
| Pasteur | x3 | ~~ | x4 | 0.546 | -0.046 | -0.072 |
| Grant-White | speed | =~ | x6 | 0.525 | -0.083 | -0.050 |
| Pasteur | speed | =~ | x3 | 0.522 | 0.149 | 0.076 |
| Grant-White | x4 | ~~ | x9 | 0.510 | 0.034 | 0.083 |
| Pasteur | x8 | ~~ | x9 | 0.498 | -0.109 | -0.185 |
| Grant-White | x4 | ~~ | x7 | 0.493 | 0.034 | 0.079 |
| Grant-White | x1 | ~~ | x2 | 0.493 | -0.073 | -0.090 |
| Pasteur | textual | =~ | x7 | 0.486 | 0.076 | 0.066 |
| Grant-White | x2 | ~~ | x7 | 0.446 | -0.047 | -0.064 |
| Grant-White | x2 | ~~ | x4 | 0.439 | 0.039 | 0.072 |
| Pasteur | textual | =~ | x2 | 0.388 | -0.086 | -0.066 |
| Pasteur | x1 | ~~ | x6 | 0.361 | 0.031 | 0.104 |
| Pasteur | x5 | ~~ | x8 | 0.337 | -0.033 | -0.069 |
| Pasteur | x3 | ~~ | x8 | 0.321 | 0.040 | 0.057 |
| Grant-White | x5 | ~~ | x8 | 0.317 | 0.028 | 0.069 |
| Grant-White | textual | =~ | x1 | 0.310 | 0.083 | 0.070 |
| Pasteur | textual | =~ | x9 | 0.247 | 0.049 | 0.047 |
| Grant-White | x6 | ~~ | x9 | 0.222 | -0.024 | -0.051 |
| Grant-White | visual | =~ | x5 | 0.199 | -0.058 | -0.039 |
| Pasteur | x5 | ~~ | x9 | 0.199 | 0.026 | 0.048 |
| Grant-White | x1 | ~~ | x8 | 0.173 | 0.028 | 0.052 |
| Grant-White | x1 | ~~ | x4 | 0.164 | 0.023 | 0.049 |
| Grant-White | x6 | ~~ | x8 | 0.140 | -0.018 | -0.045 |
| Grant-White | textual | =~ | x7 | 0.137 | 0.033 | 0.031 |
| Grant-White | visual | =~ | x6 | 0.083 | 0.037 | 0.025 |
| Pasteur | speed | =~ | x4 | 0.068 | 0.038 | 0.019 |
| Grant-White | x2 | ~~ | x6 | 0.061 | -0.015 | -0.025 |
| Pasteur | speed | =~ | x1 | 0.041 | -0.066 | -0.033 |
| Pasteur | x6 | ~~ | x7 | 0.035 | 0.009 | 0.019 |
| Grant-White | textual | =~ | x3 | 0.029 | 0.023 | 0.022 |
| Grant-White | x3 | ~~ | x8 | 0.028 | -0.010 | -0.021 |
| Grant-White | visual | =~ | x4 | 0.023 | 0.019 | 0.013 |
| Grant-White | x1 | ~~ | x6 | 0.020 | -0.009 | -0.016 |
| Grant-White | x3 | ~~ | x4 | 0.020 | -0.007 | -0.017 |
| Grant-White | x4 | ~~ | x5 | 0.019 | 0.017 | 0.048 |
| Grant-White | x4 | ~~ | x6 | 0.018 | -0.016 | -0.045 |
| Grant-White | x6 | ~~ | x7 | 0.016 | -0.007 | -0.013 |
| Grant-White | x2 | ~~ | x8 | 0.014 | -0.008 | -0.013 |

Table 5 (Modification indices)

| Group | Left | Relation | Right | MI | EPC | Standardised EPC |
|---|---|---|---|---|---|---|
| Pasteur | x2 | ~~ | x5 | 0.008 | 0.007 | 0.009 |
| Pasteur | x6 | ~~ | x9 | 0.008 | -0.004 | -0.009 |
| Grant-White | x2 | ~~ | x9 | 0.004 | 0.004 | 0.006 |
| Pasteur | x2 | ~~ | x8 | 0.002 | 0.004 | 0.004 |
| Grant-White | x1 | ~~ | x5 | 0.001 | -0.001 | -0.003 |
| Grant-White | x5 | ~~ | x6 | 0.000 | -0.000 | -0.001 |

Note. Modification indices concern fixed-to-zero parameters, not equality constraints. Add a theoretically justified parameter, then rerun the model. An addition applies only to the indicated group.

Table 6: Reliability & validity

| Group | Factor | Composite $\omega$ | AVE |
|---|---|---|---|
| Pasteur | visual | 0.584 | 0.382 |
| Pasteur | textual | 0.877 | 0.707 |
| Pasteur | speed | 0.620 | 0.351 |
| Grant-White | visual | 0.661 | 0.400 |
| Grant-White | textual | 0.878 | 0.706 |
| Grant-White | speed | 0.766 | 0.521 |

Note. Unit-weight omega-type composite reliability and AVE use observed variances (semTools). HTMT2 uses absolute correlations and geometric means; ordinal indicators use polychoric correlations. Reverse-code items in the source data when appropriate. semTools 0.5.10.

Table 7: Discriminant validity

| Group | Factor 1 | Factor 2 | HTMT2 | Latent r | $\sqrt{\text{AVE}_1}$ | $\sqrt{\text{AVE}_2}$ | Fornell–Larcker |
|---|---|---|---|---|---|---|---|
| Pasteur | visual | textual | 0.313 | 0.484 | 0.618 | 0.841 | Yes |
| Pasteur | visual | speed | 0.292 | 0.299 | 0.618 | 0.593 | Yes |
| Pasteur | textual | speed | 0.337 | 0.325 | 0.841 | 0.593 | Yes |
| Grant-White | visual | textual | 0.513 | 0.541 | 0.632 | 0.840 | Yes |
| Grant-White | visual | speed | 0.420 | 0.523 | 0.632 | 0.722 | Yes |
| Grant-White | textual | speed | 0.347 | 0.336 | 0.840 | 0.722 | Yes |

Note. Fornell–Larcker compares each factor's square root of AVE with the absolute fitted latent correlation. "Yes" means both square roots exceed that correlation; it is not a general validity verdict.